

Phase 2 Insight Report

Business Insights, Statistical Findings & Recommendations

Team: Event Horizon

Competition: Data Vortex 2026

Phase 2: SQL Analysis & Insight Generation

1. Executive Summary

This report summarizes the analytical findings from Phase 2 of Data Vortex 2026. Using a SQLite relational database built from Phase 1 cleaned data, we executed structured queries across three difficulty tiers (Easy, Medium, Hard) and two bonus analyses. The dataset contains 1,501 users and 12,000 posts across 5 social media platforms spanning May 2024 to April 2025.

2. Formula Transition (Phase 1 to Phase 2)

Phase 1 Engagement Score: $0.5 * \text{Likes} + 0.3 * \text{Shares} + 0.2 * \text{Comments}$

This weighted formula was designed to penalize click-farm inflation by reducing the weight of easily-gamed metrics (likes).

Phase 2 Total Engagement: $\text{Likes} + \text{Shares} + \text{Comments}$

We transitioned to a simple unweighted sum to strictly adhere to the client's Phase 2 definition of engagement. This ensures transparency and reproducibility.

3. Key Findings

3.1 Platform Engagement (E3)

Average total engagement by platform (filtered: 9,795 posts):

Platform	Posts	Avg Likes	Avg Shares	Avg Comments	Avg Total
YouTube	1,990	2,505.18	1,017.40	504.32	4,026.89
Instagram	1,893	2,477.70	1,044.44	501.72	4,023.86
Facebook	2,004	2,504.33	987.70	504.16	3,996.19
Reddit	1,952	2,472.87	1,000.62	510.66	3,984.15
Twitter	1,956	2,431.11	1,010.32	505.15	3,946.58

Finding: YouTube leads with 4,026.89 avg engagement per post. However, the spread across all 5 platforms is remarkably narrow (range: 80.31 points, or ~2% variance). No single platform dramatically outperforms the others.

3.2 Geographic Engagement (M1)

Top 10 locations by total engagement (filtered: 11,491 posts):

Rank	City	Country	Posts	Total Engagement	Avg/Post
1	Munich	Germany	439	1,753,727	3,994.82
2	Los Angeles	USA	438	1,750,522	3,996.63
3	Shanghai	China	437	1,735,013	3,970.28
4	Barcelona	Spain	423	1,713,206	4,050.13
5	Dubai	UAE	409	1,648,772	4,031.23
6	Melbourne	Australia	400	1,612,581	4,031.45
7	Houston	USA	397	1,565,090	3,942.29
8	Osaka	Japan	379	1,546,381	4,080.16
9	Mumbai	India	391	1,541,509	3,942.48
10	Rio de Janeiro	Brazil	390	1,534,666	3,935.04

Finding: Munich, Germany ranks #1. Crucially, engagement is NOT tied solely to English-speaking hubs. Non-English-primary regions (Tokyo, Paris, Sao Paulo, Shanghai) generate engagement sums equal to or greater than New York or London. This suggests a globally distributed audience.

3.3 Outlier Analysis (H3)

The literal query for posts exceeding 2x their platform's average engagement returns 0 rows. Diagnostic analysis proves this is mathematically correct:

Platform	Avg Engagement	2x Threshold	Max Engagement	Max/Mean Ratio
YouTube	4,026.9	8,053.8	7,755	1.926
Instagram	4,023.9	8,047.7	7,893	1.962
Facebook	3,996.2	7,992.4	7,764	1.943
Reddit	3,984.1	7,968.3	7,793	1.956
Twitter	3,946.6	7,893.2	7,628	1.933

The dataset's bounded uniform distribution (Likes max 5000, Shares max 2000, Comments max 1000) means the theoretical max engagement is 8,000. With means near ~4,000, the 2x threshold (~8,000) is unreachable. The best ratio achieved is 1.962 (Instagram).

Relative analysis (top 1% by PERCENT_RANK) identifies ~100 posts as platform-relative outperformers, with ratios ranging from 1.773 to 1.962.

3.4 Trend Analysis (Bonus)

Month-over-month engagement growth shows high volatility across all platforms:

- Instagram shows the highest single-month spike: +38.06% (Feb to Mar 2025)
- Twitter shows the deepest single-month drop: -19.0% (Aug to Sep 2024)
- YouTube shows the most stable pattern with the smallest average MoM swings

No platform shows a consistent upward or downward trend over the 12-month period, suggesting seasonal oscillation rather than structural growth/decline.

3.5 Data Anomalies (Bonus)

Anomaly classification of flagged posts:

- Missing Platform: 1,784 posts (14.87% of dataset) - the dominant anomaly
- Negative Likes: 509 posts (4.24%) - imputation artifacts from Phase 1
- HTML/Entity Corruption: Present in text_content (e.g., & entities)

All anomalies were preserved in the database and handled via SQL-level filtering rather than destructive ETL, ensuring analytical transparency.

4. Business Recommendations

4.1 Platform Strategy

YouTube and Instagram lead engagement averages, but the margin is slim (~2%). Recommendation: Do not over-invest in a single platform. Diversify content across all 5 platforms. Growth must come from increasing post volume rather than chasing viral outliers, since the data proves viral-level engagement (>2x average) does not exist in this ecosystem.

4.2 Geographic Targeting

Engagement is globally distributed. Munich, Shanghai, Barcelona, and Dubai all outperform traditional English-speaking tech hubs. Recommendation: Allocate marketing budget to non-English markets. Localized content strategies could amplify already-strong engagement in these regions.

4.3 Data Quality Pipeline

14.87% of platform data is missing and 4.24% of likes are corrupted. Recommendation: Implement upstream data validation at ingestion to prevent NULL platforms and negative engagement values. SQL-level filtering (IS NOT NULL, likes >= 0) must be mandatory for all future analytical pipelines on this dataset.

5. Assumptions & Limitations

1. Uniform Distribution Limitation: The data generation process capped maximum interactions (Likes: 0-5000, Shares: 0-2000, Comments: 0-1000). Statistically extreme outliers (like a post receiving 100x average engagement) physically do not exist. This constrains the H3 analysis.
2. Negative Likes Assumption: We assumed negative likes are data-corruption artifacts rather than 'dislikes', and filtered them from aggregate math to prevent deflation of true engagement averages.
3. NULL Platform Posts: Posts with missing platform were excluded from platform-specific analysis but included in geographic analysis (M1), since geographic engagement is platform-independent.
4. Engagement Formula: Phase 2 uses unweighted sum (Likes + Shares + Comments) per client specification. Phase 1's weighted formula ($0.5L + 0.3S + 0.2C$) is not used.
5. Time Range: Data spans May 2024 to April 2025 (12 months). Trend analysis may not capture longer-term structural patterns.

6. Methodology Summary

Database: SQLite 3 with strict relational schema, foreign keys, and CHECK constraints.

Data Load: Python script (load_data.py) with city/country splitting and imputation flag preservation.

Validation: 8-point integrity check confirming zero duplicates, correct row counts, and known anomaly preservation.

Analysis Tool: Raw SQL executed against SQLite with results exported to CSV.

Filtering Philosophy: Dynamic SQL-level filtering (WHERE clauses) instead of destructive ETL. NULL propagation in engagement calculations rather than COALESCE to zero.